Tute 7

COMP1521 26T3

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content

- 2's complement
- IEEE-754
- streams, read/write syscalls

2's complement

Decode

if msb is 0, same as normal binary
if msb is 1, the number is negative negate and add 1, then convert

Encode

convert as normal to binary
if positive -> done

Example

-13 to 2's complement binary:

13 is 0b00001101

(~0b00001101) + 1 ==

(0b11110010) + 1 ==

(0b11110011)

Then apply again to decode:

(~0b11110011) + 1 ==

(0b00001100) + 1 ==

(0b00001101) -> -13

IEEE-754

$$(-1)^{sign} \times (1 + frac) \times 2^{exp-127}$$

exp the 8 bits following the sign bit (as an unsigned 8 bit int 0..255)

frac the last 23 bits, as negative powers of 2. So 2^{-1}, 2^{-2}, ...

Good IEE-754 Calculator: www.h-schmidt.net/FloatConverter/IEEE754.html